KIMBERLY (KIMI) YASHAR
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EDUCATION

- Carnegie Mellon University, Pittsburgh, PA – Class of 2030
  - Incoming Freshman at the Mellon College of Science
  - Planning to double-major in Physics (Applied track)
- The Harker School, San Jose, CA – Class of 2026
- Relevant Completed Coursework: Honors Neural Networks, Honors Nanoscience, Differential Equations, Multivariable Calculus, AP Physics, Honors Data Structures, Honors Biology, Honors Chemistry
- Test Scores: SAT 1500
- Overall GPA: 4.37/4.8 (weighted)

SKILLS

- Software: Mathematica and Fusion 360 CAD
- Computer programming with Java
- Proficient with AI tools
- Web and graphics design
- Lab experience: Chemical etching, microscopy and imaging
- Literature review and scientific writing

RESEARCH AND ENGINEERING EXPERIENCE

- **Materials Science Researcher and Head of Microscopy (May 2025 - May 2026)**
  Starostina Lab, Aspiring Scholars Directed Research Program
  - Coauthored copper etchants paper accepted at IEEE (title: "Chemical Etchants for Dihedral Angle Measurements of Twin Boundaries in Pure Copper")
  - Created and taught a 24-student course on microscopy, optical interferometry, and CAD.
  - Tested multiple copper etchants including HNO_3, $FeCl_3$, and HCl solutions with copper and zinc.
  - Measured interfacial free energy ratio of pure Cu using Keyence Optical Interferometer.
  - Presented findings to 500+ people at Research Expos and Colloquia, including MIT Undergraduate Research Technology Conference (October 2025).

- **Optics Researcher @ University of Cambridge (October 2024 - September 2025)**
  Alban-Paccha Lab
  - Analyzed the performance tradeoffs of three optical interconnect modulator technologies (VCSELs, MOS-capacitor, and plasmonic Mach-Zehnder) across data rate, bandwidth, energy-per-bit, and voltage-length product.
  - Selected as one of eight high school students invited to present at the Chinese American Semiconductor Professional Association (CASPA) banquet for this research (September 2025).

- **Biophysics Summer Researcher @ Keck Graduate Institute (June 2024 - July 2024)**
  Niemz Lab and Casper Lab
  - Learned next generation sequencing (NGS) procedures and industry style project evaluation methods as the only high school student in two graduate research labs.
  - Designed a procedure to improve genetic screenings for pregnancies;
    - Project title: "*Non-Invasive Prenatal Testing for Detecting Congenital Heart Defects Using Next-Generation Sequencing."*
  - Performed industry-standard cost and risk evaluation reports for repurposing a drug for alcoholism that is currently used to cure lyme disease;
    - Project title: "Evaluating *Disulfiram* as a Therapeutic Candidate for *Lyme-borreliosis*: Biological and Economic Insights."
  - Selected from my research groups to present to two committees of KGI investors and professors.

- **Personal Engineering Invention (September 2022 - 2023)**
  Friction reducing hairbrush, *patents.justia.com/patent/20240138539*
    - Engineered a friction reducing hairbrush that forms air bearings around each bristle.
    - Built multiple prototypes, engineering custom circuits and mechanisms to tune its performance.
    - Applied for a patent now pending 2nd Office Action.

LEADERSHIP

- **Founder, Community Tables Cooking Classes (August 2024 - Present)**
  kimiyashar.wixsite.com/impact/communitytables
    - Founded a monthly hands-on cooking class at Sunday Friends, a food distribution and family services center in East San Jose, CA with 1,000+ members.
    - Taught 300+ families thus far in monthly 4 hour classes (in both Spanish and English).
    - Currently training 5 students to continue and expand the Community Tables program.

- **Founder, Discovery Day (February 2025 - Present)**
  kimiyashar.wixsite.com/impact/discoveryday
    - Founded an annual day of science workshops at Peterson Middle School to counter school budget cuts in the science department.
    - Designed a hands-on curriculum and led science classes for 193 students (40% English 2nd language) featuring chemistry and physics labs.
    - Now training younger members to expand the program to schools who faced similar funding cuts.

- **President, National Charity League (April 2023 - May 2025)**
    - Coordinated philanthropy, culture, and cross-grading bonding events for a chapter of 140+ mothers and daughters, contributing 800+ service hours to shelters, food banks, and education programs.
    - Raised $27,000 over 6 years for Walk to End Alzheimer's through city-wide fundraising campaigns;
        - Youngest volunteer ever awarded Elite Grand Champion for fundraising efforts.
        - Spoke with 500+ people about WTEA and my experience growing up around patients at weekly farmers market booths.

HONORS AND AWARDS

- National Merit Finalist, Less than 1% of all U.S. high school seniors - 2026
- STEM Merit Scholarship (5% of Applicants)
    - Awarded by University of Cambridge, Center of International Research for a year-long optical physics research project
- AP Scholar Award with Distinction - Highest AP testing achievement, 2025
- Presidential Volunteer Service Award (National): Gold - 2022, 2023, 2024
- California DECA – March, 2025
    - 2nd out of 117 Competitors - Integrated Marketing Campaign
    - 4th out of 94 Competitors - Business Services Marketing Role-play
- Harker DECA – Experienced Leader Award (1 of 189 members) - 2024
    - Awarded to a member of the chapter who best exemplifies leadership and mentorship skills
- $1,000 Inspire Innovation Grant Recipient - August, 2024
    - Awarded by National Charity League to one applicant whose initiative is impactful and transformative